

1590 First-in-human study of ES014, a bispecific antibody targeting CD39 and TGF- β as monotherapy in patients with advanced solid tumors

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Background: ES014 is a first-in-class bifunctional antibody-ligand trap that targets the adenosine and TGF- β pathways. By simultaneously targeting CD39 and TGF- β , ES014 can inhibit the production of immunosuppressive adenosine, promote the formation of immunostimulatory ATP, and neutralize immunosuppressive cytokine TGF- β . We conducted a phase I study to evaluate the safety, tolerability, PK and preliminary antitumor activity of ES014 in patients (pts) with advanced solid tumors.

Methods: This is a phase I, open-label, multicenter study with two parts: a dose escalation part and a cohort expansion part. Eligible pts were defined as advanced solid tumor pts with progressive disease after standard therapies. Baseline tumor CD39 expression was assessed by IHC staining and scored using the combined positive score (CPS).

Results: As of July 08, 2025, 75 pts received ES014 treatment. Among the 75 pts, the median age was 62.2 years. 54 (72%) were males. No DLT was observed. Twenty pts (26.7%) experienced $\geq$ Grade 3 TEAEs. The overall incidence of TRAEs that resulted in drug discontinuation was low (4%). No patient had a TRAE that resulted in death. The most common TRAEs were anaemia, rash, and pruritus. Among the 19 evaluable non-small cell lung cancer (NSCLC) pts treated at the recommended doses who had high CD39 expression at baseline (CD39 CPS $\geq$ 5), 1 achieved PR and 10 had SD, resulting in a disease control rate (DCR) of 57.9%. Among the 10 gastrointestinal stromal tumor (GIST) pts, 1 achieved PR and 6 had SD, resulting in a DCR of 70%. Among the 5 desmoid tumor (DT) pts, 1 achieved PR and 4 achieved SD, resulting in a DCR of 100%. One NSCLC patient remained on study for over 50 weeks. The systemic exposure of ES014 increased more than dose-proportionally between 20 and 1400 mg with no accumulation. A receptor occupancy of over 90% was achieved at dose levels $\geq$ 700 mg and the inhibition of the total peripheral TGF- β was over 80% at dose levels $\geq$ 70 mg.

Conclusions: ES014 demonstrated a good safety, tolerability, PK and PD profile in pts with advanced solid tumors. ES014 monotherapy showed preliminary antitumor activity in various tumor types including NSCLC, GIST and DT. There appears to be a correlation between CD39 expression and clinical response.

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160MO BL-M07D1, a novel anti-HER2 antibody-drug conjugate (ADC), in subjects with metastatic HER2-mutant non-small cell lung cancer (NSCLC)

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Background: BL-M07D1 is an anti-HER2 antibody drug conjugate (ADC) comprised of a humanized anti-HER2 antibody, a cathepsin B cleavable linker, and a novel topoisomerase I inhibitor (Ed-04).

Methods: BL-M07D1-201 is a single arm, multicenter, phase II study evaluating the safety and efficacy of BL-M07D1 in previously treated patients (pts) with HER2-mutant (HER2mut) NSCLC. All enrolled pts were treated at 3.8, 4.4, 5.0mg/kg D1Q3W.

Results: As of July 12, 2025, 58 pts were enrolled at 3.8 (n=5), 4.4 (n=50), and 5.0 mg/kg (n=3). All pts who received at least one dose of BL-M07D1 are included in the analysis. The median follow-up was 7.9 months. For 50 pts treated at 4.4 mg/kg, the ORR was 62.0%, cORR was 54.0%, mDOR was not reached with a 6-mo DOR rate of 95.7%, and mPFS was not reached with a 6-month PFS rate of 90.7%.

Table: 160MO

	4.4 mg/kg D1Q3W (N = 50)
Median prior treatment line (range)	1 (1-7)
Best overall response	
PR	31
PR confirmed	27
PR pending confirmation	2
SD	17
PD	0
NE	2
ORR, % (95% CI)	62.0 (47.2, 75.3)
cORR, % (95% CI)	54.0 (39.3, 68.2)
DCR, % (95% CI)	96.0 (86.3, 99.5)
mDOR, mo (95% CI)	NR (6.9, NR)
6-mo DOR rate, % (95% CI)	95.7 (72.9, 99.4)
mPFS, mo (95% CI)	NR (9.8, NR)
6-mo PFS rate, % (95% CI)	90.7 (76.9, 96.4)

The most common TRAEs ($\geq$ 20%, all grade/ $\geq$ G3) were anemia (79.3%/31.0%), thrombocytopenia (75.9%/37.9%), leukopenia (60.3%/34.5%), neutropenia (46.6%/36.2%), nausea (46.6%/3.4%), asthenia (44.8%/6.9%), alopecia (43.1%/0), decreased appetite (36.2%/0), weight decreased (36.2%/0), ALT increased (27.6%/0), hypoalbuminemia (25.9%/0), lymphocyte count decreased (24.1%/6.9%), vomiting (22.4%/0), and diarrhea (22.4%/5.2%). Grade 3 and above TRAEs which were predominantly hematologic in nature, were able to be effectively managed with standard supportive measures including dose reductions, as demonstrated by the TRAE leading to discontinuation rate of 3.4%. One death due to cardio-respiratory arrest was reported possibly related to BL-M07D1 assessed by investigator. No ILD was observed, and no new safety signals were identified.

Conclusions: BL-M07D1 demonstrated encouraging efficacy with a manageable safety profile in previously treated pts with HER2mut NSCLC.